

The Isolated and Perfused Brain of the Guinea-pig *In Vitro*

M. Mühlethaler¹, M. de Curtis², K. Walton³ and R. Llinás³

¹Laboratoire de Psychophysiologie, Département de Physiologie, CMU, 1 Rue Michel Servet, 1211 Genève 4, Switzerland

²Present address: Dipartimento di Neurofisiologia, Istituto Neurologico, via Celoria 11, 20133 Milano, Italy

³Department of Physiology & Biophysics, New York University Medical Center, 550 First Avenue, New York, NY 10016, USA

Key words: visual system, superior colliculus

Abstract

We describe here an isolated and perfused *in vitro* adult guinea-pig whole brain preparation which is an extension of the previously described *in vitro* brainstem–cerebellum preparation. Viability was tested by the analysis of trans-synaptic responses along the visual pathways following the electrical stimulation of the optic nerve or the optic radiations. The evoked field potentials were recorded in the dorsal lateral geniculate, the superior colliculus and the visual cortex. The distribution of extracellular currents was studied using current source density analysis, in order to determine the amplitude, time course and spatial organization of the synaptic activity at these sites. The study indicates that field potentials were very similar to those described *in vivo*. These data demonstrate the survival of a complex adult sensory system *in vitro* and suggest that this preparation can be used for the analysis of multisynaptic circuits in the mammalian brain.

Introduction

The technical difficulties involved in obtaining stable intracellular recordings and in accessing deep brain structures under *in vivo* conditions stimulated the development of alternative *in vitro* techniques (reviewed in Grantyn and Kettermann, 1992). Of these the most useful and popular one thus far has been the brain slice. The early limitations of this preparation for certain types of studies prompted the development of techniques for recording from acutely dissociated single mammalian brain neurons (Kay and Wong, 1986).

In addition to these 'simplified' preparations, it has also become clear that more complex *in vitro* preparations were required, ones in which multisynaptic circuits and the intracellular activity they generate were well preserved. Electrophysiological studies using preparations comprising large blocks of CNS tissue were first developed using the nervous system of lower vertebrates (Hackett and Llinás, 1973). As far as mammalian tissue is concerned, perfused tissue blocks of the brainstem, the brainstem–cerebellum or the basal hypothalamus were successfully introduced in the early and middle eighties (Llinás *et al.*, 1981; Walton and Llinás, 1982; Bourque and Renaud, 1984; Llinás and Mühlethaler, 1988a,b). In addition the superfused spinal cord of young rats (Otsuka and Konishi, 1974; Bagust *et al.*, 1985; Fulton and Walton, 1986) have also been successfully utilized.

The feasibility of maintaining an isolated, adult mammalian brain *in vitro* remained to be determined. The arterially perfused *in vitro* adult guinea-pig brain preparation was introduced a decade ago (Walton and Llinás, 1982), but it was only following the development of better perfusion techniques with the reduced isolated and perfused brain-

stem–cerebellum (Llinás and Mühlethaler, 1988a,b) that the possibility of maintaining the entire brain *in vitro* was reassessed.

The present study addresses the question of the functional integrity of the extended neuronal circuits and of monosynaptic as well as polysynaptic interactions at different levels of the neuraxis in the isolated adult guinea-pig brain. For that purpose it was determined that the study of field potentials, as in the brainstem–cerebellum (Llinás *et al.*, 1981; Llinás and Mühlethaler, 1988a,b), was the most sensitive way to study the functional integrity of large populations of neurons. Indeed, single-cell studies reflect the viability of individual elements, but not that of the larger percentage of the neuronal population. Here we have chosen to study the visual system since its anatomy and function is well known and it can be investigated in great detail using field potential analysis at several levels following optic nerve stimulation. Given that such field potential analysis has been carried out using *in vivo* preparations by many investigators (Freeman and Singer, 1983; Mitzdorf and Singer, 1977, 1978) it provides a clear standard for determining the viability of the isolated *in vitro* brain.

This paper provides the first detailed and complete account of the techniques developed to isolate and maintain a perfused adult guinea-pig brain preparation *in vitro*. Several papers have appeared using this technique to study various regions of the brain, such as the entorhinal cortex (Alonso *et al.*, 1990), thalamus (Mühlethaler and Serafin, 1990) and limbic system (de Curtis *et al.*, 1991; Paré *et al.*, 1992). A number of abstracts have also appeared reporting intracellular or extracellular studies of the cerebral cortex (Walton and Llinás, 1982; de Curtis and

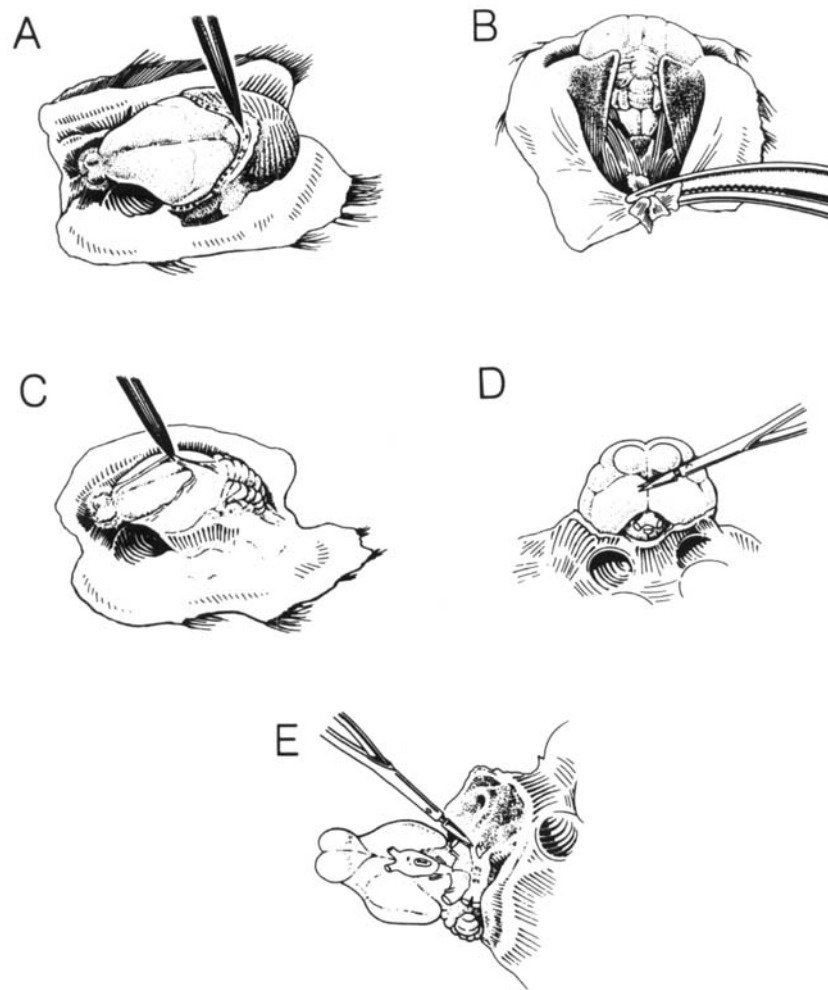


FIG. 1. Steps in the isolation of the guinea-pig brain. (A) Dorsal view of the brain after removal of the overlying frontal and parietal bones. (B) Posterior view of the brain after removal of occipital bone and section of the spinal cord. (C) Removal of the dura from the surface of the brain. (D) Lifting the brain to expose the optic nerves and internal carotid arteries. (E) Section of the cranial nerves after the brain has been tilted back.

Llinás, 1989; Llinás *et al.*, 1989; Mühlethaler and Macvicar, 1989), facial nuclei (Mühlethaler *et al.*, 1987), gigantocellular reticular formation (Serafin and Mühlethaler, 1988), hypothalamus (Khateb *et al.*, 1990) and limbic system (de Curtis and Llinás, 1989; Paré and Llinás, 1991).

Materials and methods

The technique used for the isolated whole brain is a modification of that previously described for the isolated brainstem/cerebellum (Llinás and Mühlethaler, 1988a,b).

Dissection

Experiments were performed on young adult guinea-pigs weighing 150–200 g. General anaesthesia (Nembutal, 30–60 mg/kg i.p.) was followed by rapid thoracotomy and transcardiac perfusion of saline through the ascending aorta. At this point the right auricle was cut and the descending aorta clamped. Following blood washout by a few seconds of perfusion with a room temperature (22–23°C) solution, a 10°C solution was perfused for 5–6 min to rapidly lower the temperature of the brain. To hasten the cooling, the head of the animal

was placed in ice-cold water. The perfusate was fed by gravity, giving a flow rate of 30–50 ml/min and a perfusion pressure of 80–100 mm Hg. It was particularly useful for the transcardiac perfusate to pass through a bubble trap just before entering the heart so that any bubbles which may form by the warming of the perfusate did not enter the circulatory system. At the end of the perfusion the temperature of the brain was near 10°C.

Once the brain had reached a steady temperature, transcardiac perfusion was stopped, the catheter rapidly withdrawn and the head was severed. The skin over the skull as well as the periosteum were dissected away. Using a small bone rongeur, the nasal and frontal bones were then removed, exposing the olfactory bulbs and frontal lobes (Fig. 1A). The tip of the rongeur was then inserted at bregma and the parietal bone on one side was removed in one swift movement. The contralateral parietal and occipital bones were then removed, revealing the rest of the cortex as well as the cerebellum (Fig. 1B). At this point the bone over the cerebellar paraflocculi was delicately disarticulated in order to avoid damaging the underlying tissue. If the transcardiac perfusion had been successfully performed, the brain would appear white, with all traces of blood removed from the arterial and nervous systems. After removing the cervical vertebral laminae, the spinal cord

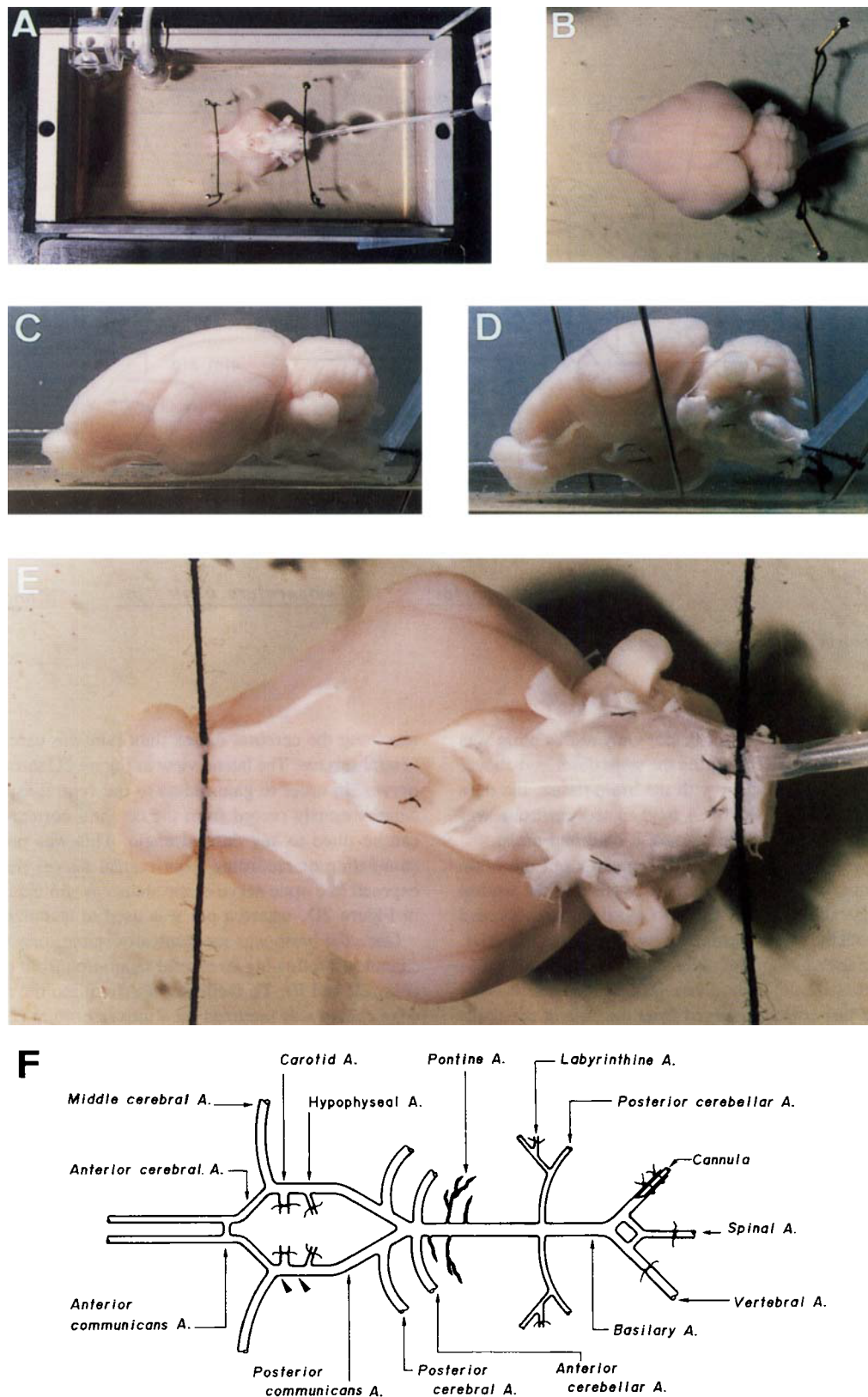


FIG. 2. Cannulation and perfusion of the brain in the incubation/recording chamber. (A–E) Views of the brain at different magnifications. (F) Simplified view of the ventral aspect of the arterial system of the guinea-pig brain. The two arrowheads indicate where, on the circle of Willis, knots should be placed if a carotid artery is damaged during isolation of the brain.

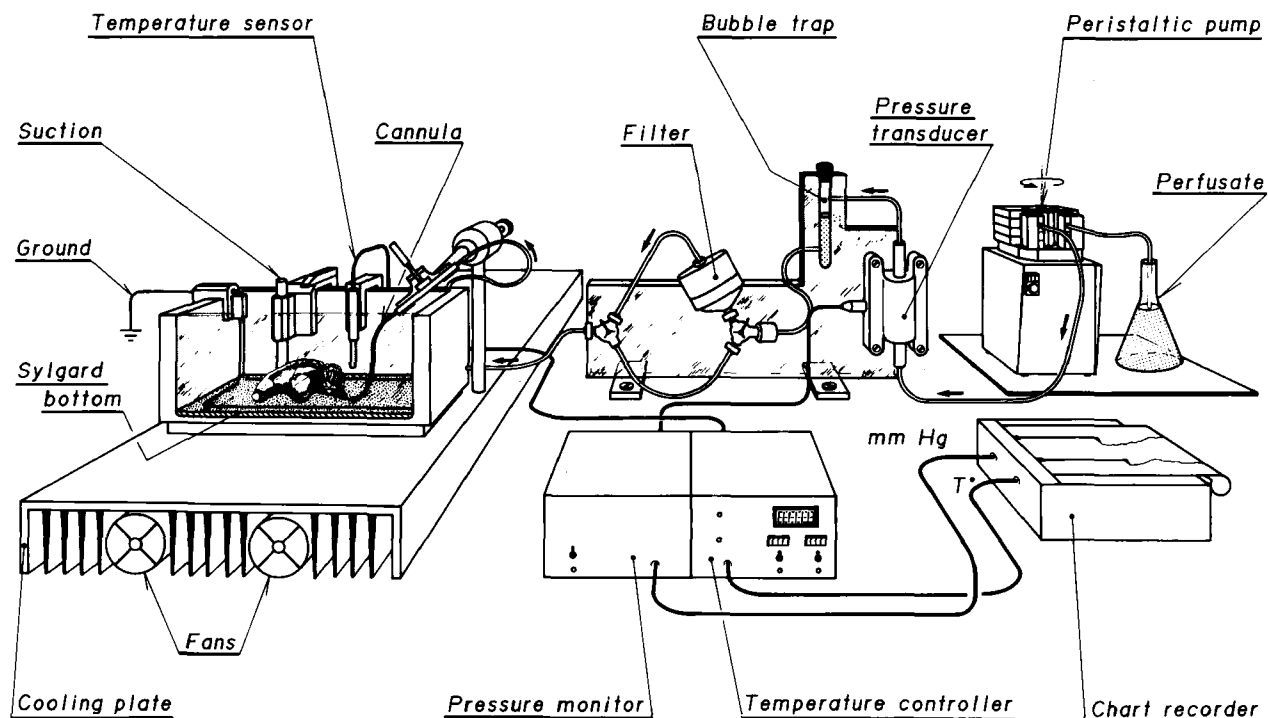


FIG. 3. Incubation/recording chamber and perfusion system (see text).

was transected at C2–C3 using a razor blade. Only at this point was the dura removed (Fig. 1C). Indeed, during the steps described above, in order to minimize harmful contact with the brain tissue, the dura was kept intact, while a cold (5–10°C) oxygenated perfusate was continuously dripped over the brain, keeping it cold and moist.

The orbital crest was then removed with rongeurs and the brain progressively tilted out of the skull by delicately pushing the ventral surface of the olfactory bulbs using a spatula wrapped with a moistened covering of cloth or with the side of a pair of surgical scissors (Fig. 1D). As the brain was separated from the skull, a dissecting microscope was used to better visualize the carotid arteries and the optic nerves; these were transected just as they emerged from the base of the skull. This part of the dissection is the most crucial and delicate since the carotid arteries can easily be torn. The pituitary stalk and the oculomotor nerves were sectioned next. Following this step the brain was progressively tilted to reveal the remaining cranial nerves as they emerged from the brainstem (Fig. 1E). After the last cranial nerves and the vertebral arteries had been transected the brain was gently immersed in cold, oxygenated saline and transferred to the recording chamber, which was maintained between 12 and 15°C. Isolation of the brain should take between 8 and 15 min (not including the 5–6 min of transcardial perfusion). We have found that careful dissection is more important than speed in preserving a viable isolated brain.

Cannulation and perfusion

The brain was placed on its dorsal surface in the chamber (Fig. 2A) and stabilized by two threads fixed to the chamber bottom, which was coated with Sylgard (Dow Corning) (Fig. 2A and E). At this time a layer of small glass beads (Cole Palmer, 0.15 mm diameter) was poured into the chamber to enfold the brain at its base. This provided additional mechanical stability to the encephalic mass as the tissue was partly buried in the bead matrix. As shown in Figure 2B, for experiments

involving the cerebral cortex the brain was carefully turned onto its ventral surface. The lateral view in Figure 2C shows the severed cranial nerves. In order to gain access to the ventral aspect of the brain and simultaneously record from the cerebral cortex (Fig. 2D), the brain can be tilted to any desired angle. This was particularly useful for stimulation or recording from cranial nerves that are not otherwise exposed (the optic nerve or the abducens, for example). This is shown in Figure 2D, where a pin was used to stabilize the tilted brain.

Once the brain was mechanically stable, one vertebral artery was cannulated following its careful separation from the surrounding dura (Fig. 2E and F). To facilitate insertion into the artery, the bevelled-edge cannula was mounted on a universal joint (Figure 3) and a slow flow of perfusate (1 ml/min) delivered through the cannula in order to expand the artery during cannulation. After the cannula was tied inside the vertebral artery (Fig. 2E and F), the perfusion flow was increased to 2 ml/min and phenol red was added to the perfusate at a final concentration of 1%. Leaking dye helped to locate any torn vessels, yet did not damage the brain. The contralateral vertebral artery as well as the spinal artery were both tied as shown in Figure 2E and F.

The next step was to tie the internal carotid arteries (Fig. 2E), which were occasionally quite short. In this case several consecutive ties were required to achieve complete fluid stasis. Sometimes one of the carotid arteries had been torn and ties placed on either side of the lesioned artery (double arrowheads in Fig. 2F) effectively prevented any leaks. Circulation was completed around the Willis circle, via the anterior communicating arteries.

After the carotid arteries were tied, the hypophyseal and the labyrinthine arteries, as well as some minor brainstem arteries severed during the dissection, were also tied. The whole procedure from the time the brain was brought to the chamber to the time all the leaks were eliminated would take from 15 to 30 min (with the cannulation and the beginning of perfusion itself taking the initial 1–3 min).

The ionic composition of the perfusate was: 126 mM NaCl, 26 mM NaHCO₃, 1.2 mM KH₂PO₄, 1.3 mM MgSO₄, 3 mM KCl, 15 mM glucose, 5 mM HEPES and 2.4 mM CaCl₂. Dextran 70 (Macrodex 70, Pharmacia) was added (3%) to restore oncotic pressure. The perfusate was bubbled with a mixture of 95% O₂ and 5% CO₂ (pH 7.4). After ~30–60 min (including the cannulation and tying of the vessels) at 12–15°C with a perfusion rate of 2 ml/min, the brain was progressively rewarmed to 30°C over a period of at least 90 min. For long experiments in which pathways were tested repetitively, the temperature was kept at 28–30°C. The perfusion flow was increased in parallel, such that at 30°C it would be set to 4–5 ml/min, yielding a perfusion pressure near 80 mm Hg (see discussion in Llinás and Mühlethaler, 1988a,b). Given this time of preparation, experiments were usually started 2–3 h after the onset of the procedure. Altogether, the brain survived about 8 h in isolation; 70–80% of this time could be used for electrophysiological recording.

While the initial experiments were performed in the original chamber (Llinás and Mühlethaler, 1988a,b) an improved version is now available (Mühlethaler *et al.*, 1990) (Fig. 3). As the perfusate was moved towards the preparation by a peristaltic pump (Gilson), it passed through a flow-through pressure transducer (Honeywell), a bubble trap, and a submicron filter (Millipore, 0.22 µm pore diameter). A bypass allowed the filters to be changed without interrupting the perfusate supply to the brain. This is needed since the filter may be clogged during the experiment (particularly after perfusion with phenol red). The perfusion line then passed into the back panel of the thermoregulated chamber, where it coiled and came back out through the cannula, the temperature of the saline at that point being at the desired value. Thermoregulation of the chamber was provided by two Peltier elements located beneath the chamber. A large heat sink (cooling plate) dissipated the heat when the perfusate was being cooled. This was facilitated by two small fans below the cooling plate (Fig. 3), which were activated only when the chamber was maintained at 12–15°C. The excess heat may also be dissipated by a water circulation system. The bottom of the chamber was covered by a layer of Sylgard, and the holders for the ground wire, aspiration holder and temperature probe were fixed to the side of the chamber. The thermoregulation unit stabilized the temperature at any desired value between 10 and 37°C. This value was monitored on the chart recorder throughout the experiment, as was the perfusion pressure (Fig. 3).

Some important points

Animals

We found the ideal weight of the guinea-pigs to be between 150 and 200 g. Preparations obtained from younger animals (<150 g) are more difficult to handle due to the softer consistency of the skull and the more delicate nature of the brain tissue. In contrast, older animals have a much more resistant skull, cranial sutures and encephalic mass.

Timing

Because the brain is perfused with cold saline before the animal is decapitated, the duration of the dissection that follows is not a critical factor, within certain limits. Typically, good survival has been observed with dissection times ranging from 8 to 15 min (from the time the head is severed to the time the vertebral artery is cannulated and perfusion begun). Dissections, such as removal of the neocortex to expose the hippocampus, may be carried out during the transcardial perfusion with cold saline (Paré *et al.*, 1992).

Vessels

In contrast to the slice preparation, since the isolated brain depends exclusively on the perfusion of its arterial system for supplying oxygen

and nutrients and for removing toxic substances, if the intricate and delicate network of vessels on the ventral surface of the brain is damaged, the preparation will not survive. Major arteries (shown in Fig. 2F) are easily identified, but smaller and still important arteries on the ventral and dorsal surfaces of the brain may be easily overlooked. In addition to generating local anoxia they collectively constitute a major source of leakage and diminish the viability of the preparation. Particular care must thus be given to manipulation of the pial surface, in which a large number of small vessels are embedded. As in *in vivo* experiments, recording and stimulating electrodes cannot easily penetrate the pia matter, and rather than tearing it, a small window should be made with scissors or the tip of a scalpel to minimize disruption of the local vascularization. The flow of phenol red is, in any case, the best index for small leaks. It should nevertheless be stressed that very small leaks are always present at various sites, and may represent either venous flow or minutely lesioned arterioles.

Saline

While carefully following the procedures described above will ensure several hours of survival for the preparation, it is noteworthy that after 6–8 h the brain will usually begin to swell, although excellent recordings can still be made. It has been suggested (de Curtis *et al.*, 1991) that a free radical scavenger such as thiourea, added at 400 µM to the perfusate (Rice and Nicholson, 1991), could delay the oedema. A second possible modification of the conditions of perfusion is superfusion (at 6–10 ml/min) in the chamber with a dextran-free saline (Llinás and Mühlethaler, 1988). This is particularly useful for rapidly removing substances from the external bath during pharmacological experiments, but is apparently not necessary for the survival of the isolated brain.

Maintenance

While, as in the case of brain slices, sterile conditions are not required, we found that the survival of the brain critically depends on having a perfectly clean perfusion system. This is best achieved by using a 3% H₂O₂ solution to systematically clean the perfusion line after each experiment (except for the pressure transducer, which can eventually be damaged by this procedure). Also, it is important to change the filter as soon as the pressure increases during an experiment. The bypass in the perfusion line (Fig. 3) was introduced to allow continued flow to the brain while the filter is being changed. Extreme care should be taken, however, to avoid injecting bubbles into the perfusion line, as they block the fine capillaries and are usually detrimental to the preparation.

Extracellular recordings: field potentials and current source density analysis

Field potentials were obtained following electrical stimulation with either bipolar twisted silver wires (insulated except at the tip) or concentric bipolar electrodes. The field potentials were recorded with glass micropipettes filled with 2 M NaCl. In some experiments the location of the electrode tip was determined histologically after ejection of Alcian blue or Chicago blue (at saturated concentrations in the micropipette conductive solution). Electrode tracks were reconstructed from Alcian blue dye marks deposited by pressure ejection at the end of the penetrations and along the electrode withdrawal track. With the cannula attached, the brain was placed in a second chamber and perfused through the cannula with 10% formalin, or glutaraldehyde, and the cannula was then discarded.

Demonstration of the preservation of the intrinsic connectivity within brain structures at different levels in the neuraxis was achieved by analysing field potential profiles obtained from averaged evoked

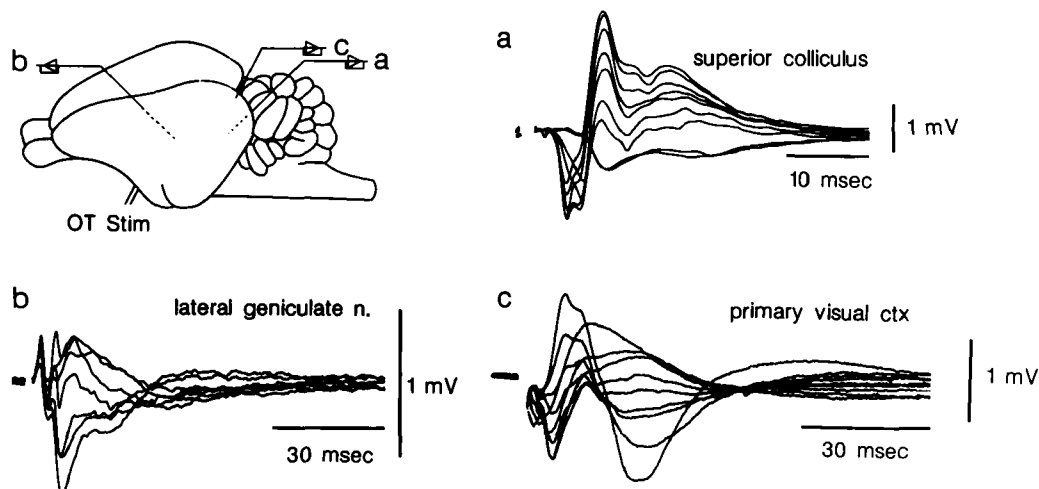


FIG. 4. Averaged evoked potentials recorded in the superior colliculus (a), dLGN (b) and visual cortex (c) following stimulation of the optic tract. The positions of the stimulating and recording electrodes are illustrated schematically in the drawing on the left. Each panel shows superimposed traces obtained at different depths along an electrode penetration perpendicular to the lamination of the structure under study.

potentials (AEPs) recorded at 50 μm steps along tracks perpendicular to the lamination of the structures under study. Field potentials were recorded using a Neurodata 384 amplifier and stored on video tape using a Neurocorder interface (Neurodata Instruments, New York). One-dimensional current source density (CSD) analysis was implemented with a Nicolet 4094 digital oscilloscope on profiles of AEPs evoked by tract or nerve stimulation. This allowed us to (i) locate the distribution of the synaptic events generated sequentially by the neural populations, (ii) examine the connectivity involved in generating the particular AEP profile, and (iii) characterize the temporal aspects of pathway conduction.

The theoretical basis of CSD analysis has been described in detail (Freeman and Nicholson, 1975; Mitzdorf, 1985). The validity of our measurements was based on the assumption that the nervous tissue behaves as an almost purely resistive medium. We also assumed that extracellular space conductivity does not change significantly along the electrode penetration (Nicholson and Freeman, 1975). Since one-dimensional CSD analysis has been carried out in the lateral geniculate (Mitzdorf and Singer, 1977), cat superior colliculus (Freeman and Singer, 1983), and visual cortex (Mitzdorf and Singer, 1978), our results could be directly compared with *in vivo* findings.

Results

The viability of the visual system was tested by recording depth field potentials following optic tract stimulation (for positions of electrodes see drawing in Fig. 4). Evoked potentials were obtained and averaged at 50 μm steps in penetrations perpendicular to the lamination of the structure. Superimposition of these AEPs illustrates the expected reversal in current flow as the microelectrode passed through the superior colliculus (Fig. 4a), dorsal lateral geniculate nucleus (dLGN) (Fig. 4b) and primary visual cortex (Fig. 4c). A detailed analysis was made of the superior colliculus and cortical field potentials as they reflect the integrity of long pathways with mono- and multisynaptic components.

The superior colliculus

Recordings from the superior colliculus were obtained along dorso-ventral penetrations perpendicular to the superior colliculus lamination

(Fig. 5A). The waveform of the extracellular field potential following optic tract stimulation changed as the electrode penetrated from the surface through the superficial layers of the superior colliculus (Figs 4a and 5B). The largest amplitude responses were recorded from the two most superficial layers, the stratum griseum superficiale (SGS) and the stratum opticum (Fig. 5A, B). The terminology introduced by Fukuda *et al.* (1978) was utilized to describe the AEP's components in this study. Because the superior colliculus was not directly accessible under visual control from the surface of the brain, the track of the recording electrode was histologically reconstructed on serial coronal sections through the tectum following the deposition of an Alcian blue spot (arrow in Fig. 5A) at the end of the penetration.

While the analysis of field potentials gives a general picture of the current flowing in the superior colliculus following its afferent activation, the location and time course of the current sources and sinks corresponding to synaptic events is best achieved through one-dimensional CSD analysis. The right column in Figure 5B shows samples of the CSD profiles calculated on sets of averaged field potentials (left column) obtained during a dorso-ventral penetration through the superior colliculus. The synaptic events were preceded by an early negativity best seen in the deeper traces (Fig. 5B, arrow) that correspond to the presynaptic input volley. The dotted lines on the CSD profiles show the latencies of the synaptic sinks responsible for the field potential components described below (Fig. 6). Four of these sinks are clearly visible in Figure 5B: an early one in the bottom trace (at 6.5 ms), a slightly delayed one (at 8 ms) in trace c and two much later ones (at 11 and 21 ms) in the two upper traces.

The change in current flow with depth is shown in Figure 6. Panel B illustrates CSD trace pairs recorded at two different depths. The contribution of the current sinks and sources generated by four separate excitatory synaptic events is illustrated by the filled pattern. Four major waves were distinguished in the field potentials and in the corresponding CSD records (Figs 5B and 6B). The first wave was a short-latency sink (N1), which reversed in sign (P1) more superficially (Fig. 6B). This event was shortly followed by a second sink (N2; 8–10 ms), which reversed in sign (P2) at the same depth as P1. The third and the fourth waves (N3; 10–13 ms and N5; 17–20 ms) were negative in the superficial layers and positive (P3 and P5) deeper within the superior colliculus. These last two components reversed in polarity more

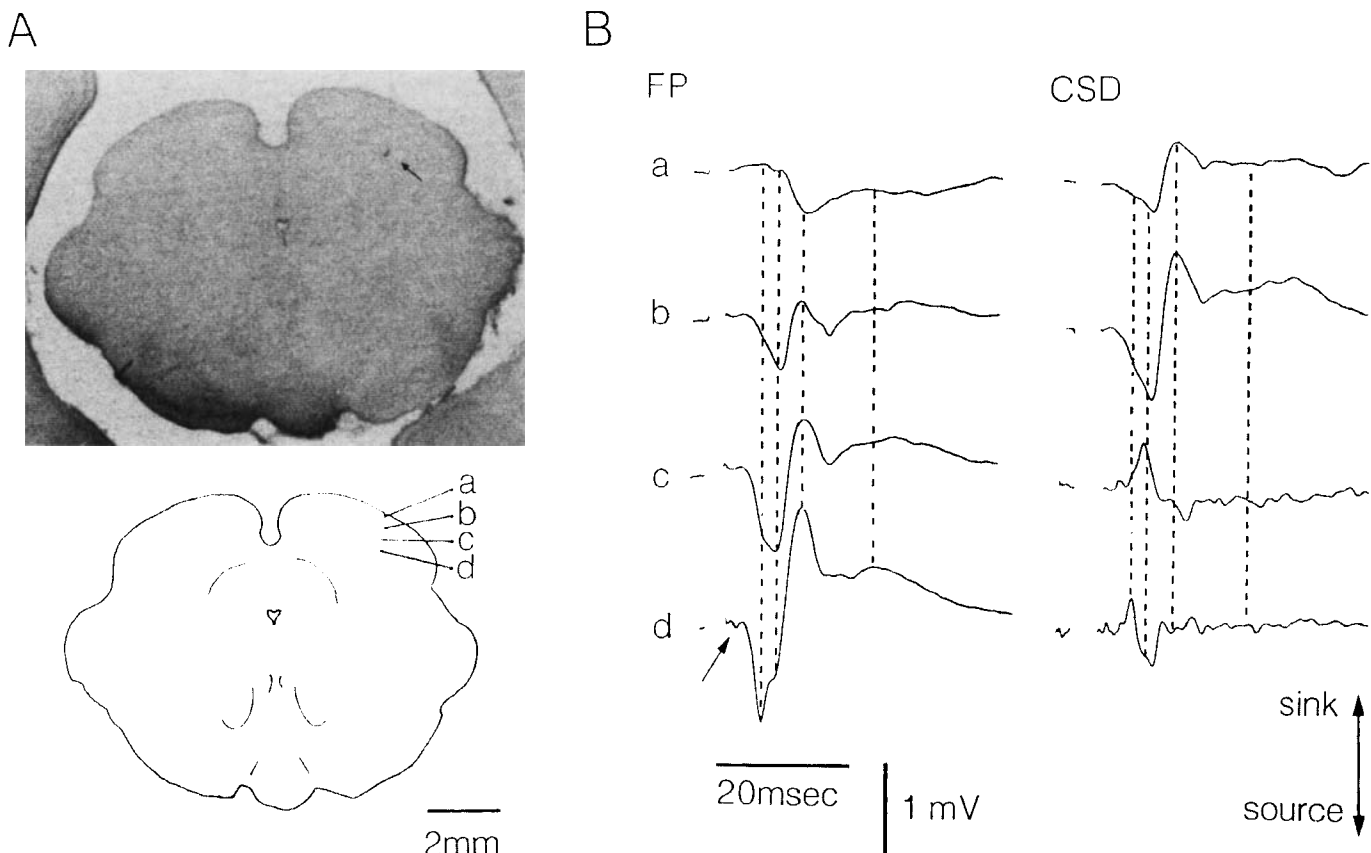


FIG. 5. (A) Microphotograph and schematic drawing of a coronal section of the brainstem at the level of the superior colliculus. The upper panel shows the location of a spot of Alcian blue (arrow), which was pressure-ejected through the recording electrode in order to reconstruct the electrode track. The drawing (bottom panel) suggests the locations at which the averaged evoked potentials illustrated in B were obtained. (B) Field potential recordings (FP) and related current source density (CSD) traces recorded from the surface (a), the border between the stratum zonale and SGS (b), in the SGS (c) and in the superficial stratum opticum (d) along the electrode penetration illustrated in A. The arrow in d shows the presynaptic volley. Broken lines indicate times (6.5, 8, 11 and 21 ms) at which the measurements shown in Figure 6 were made.

superficially than the first two waves and were largest in the upper SGS, close to the border with the stratum zonale.

The CSD depth profile (Fig. 6A) of the earliest component reveals a current source in the upper SGS and a sink near the SGS–stratum opticum border (Figs 5B and 6A,B; 6.5 ms). The current flow generating the second component of the field potential (Figs 5B and 6A, B; 8 ms) was quite similar to the first but the sink was more superficial than that at 6.5 ms. The short latencies of the sinks suggested activation of a monosynaptic input (N1 and N2 in Fig. 6C). The late, more superficial sinks at 300–400 μm corresponded to peaks N3 and N5 at 11 and 21 ms, respectively (Fig. 6A, B) [representing superior colliculus activation by fast and slow W-type fibres of the OT (N3 and N5 in Fig. 6C)]. In some penetrations a fifth sink was observed, with the same location as components N3 and N5. This sink could be ascribed to currents activated by polysynaptic intracollicular circuits. In short, the recordings agree with those obtained under *in vivo* conditions (Freeman and Singer, 1983; King and Johnson, 1985).

Visual cortex field potentials

Activity in the visual cortex clearly reflects the activation of polysynaptic pathways. Indeed, activation of the optic tract elicited complex synaptic potentials in the dLGN, the first relay nucleus for the input to the cortex. The lateral geniculate responses ($n = 7$) were characterized by two short-latency components (Fig. 4B) probably due

to monosynaptic responses following Y- and X-fibre activation respectively.

Analysis of the AEPs elicited in the primary visual cortex by optic tract stimulation or optic radiation stimulation revealed sequential activation of a polysynaptic pathway. This activity flowed through layer IV (600 μm , Fig. 7A) to the supra- and infragranular layers. The CSD analysis of optic tract-evoked field potentials (CSD, Fig. 7A, B) showed an early sink in layer IV (600 μm) 5–7 ms after the stimulus artefact. This response was most probably due to monosynaptic activation of the granular cells by the thalamic input (Fig. 7C, synapse 1). This activity was followed by two large sinks with peak amplitude at 250 μm above and 800 μm below the granule cell layer (8–10 ms). They represent the disynaptic activation of pyramidal cell dendrites in layers II–III and V, respectively (Fig. 7A, synapses 2). A late sink was then revealed in layer I (100 μm) and deep in the cortex (1000 μm); these late synaptic events were due to the trisynaptic activation of superficial and deep pyramidal neurons (13–17 ms) (Fig. 7C, synapses 3). The deep and large sources that follow the sinks at each synaptic location were probably mediated by currents activated by local circuit inhibitory interneurons. Similar results were reported by Mitzdorf and Singer (1978) in the cat *in vivo*.

The synaptic components in the AEPs and in the CSD profile are illustrated at a faster time base in Figure 7B, where samples of the profile in A are illustrated. The dotted lines mark the mono-, di- and

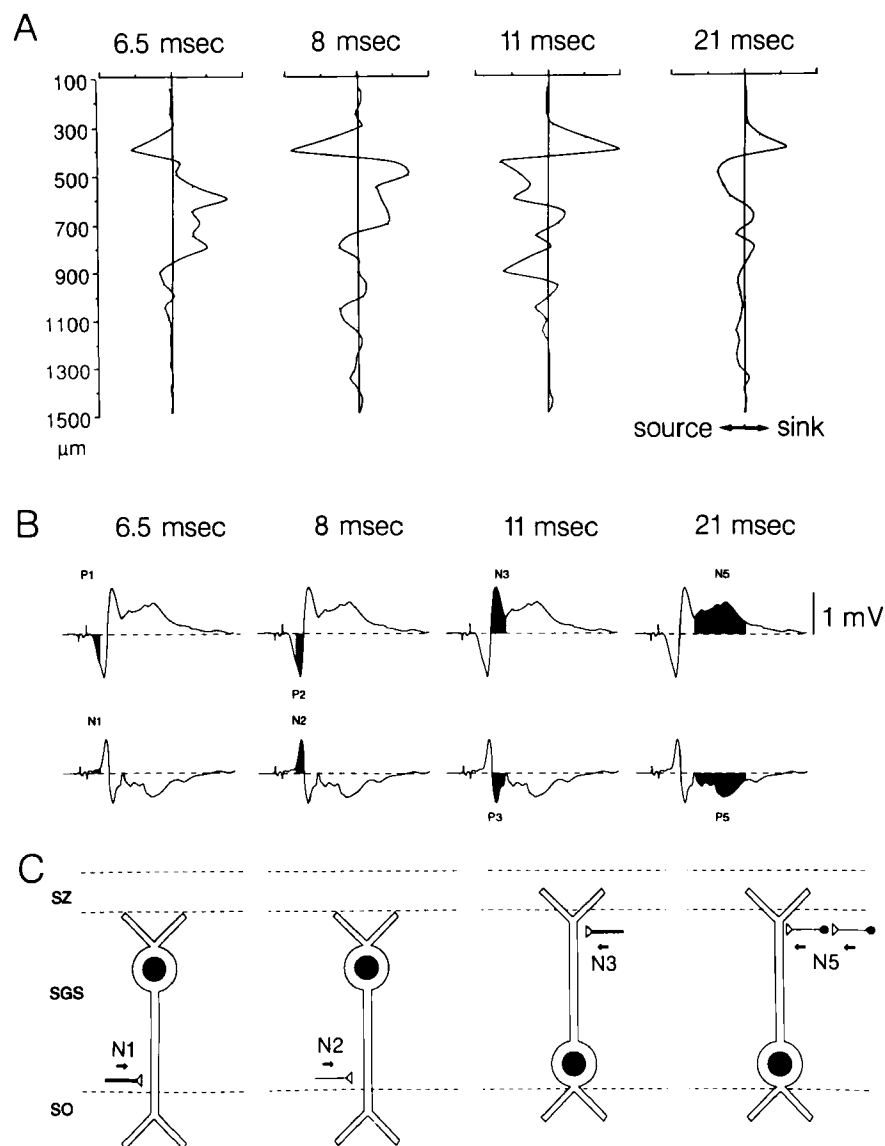


FIG. 6. (A) Current source density (CSD) depth profiles obtained at four latencies during a superior colliculus penetration. The amplitude of the CSD traces at 6.5, 8, 11 and 21 ms is plotted against the depth from the dorsal surface of the superior colliculus. Leftward and rightward deflections indicate a source and a sink, respectively. (B) Current source density of averaged evoked potentials recorded from a superficial (top row) and deep (bottom row) area of the superior colliculus. In each trace the activity at the time indicated is shaded. (C) Diagram of superior colliculus showing four cells and location of activity corresponding to shaded area of traces in B.

trisinaptic components of the cortical response at different depths along the penetration (peaks 1, 2 and 3, respectively). This multisynaptic sequential invasion of the visual cortex following peripheral stimulation of the optic pathway is summarized in the schematic drawing in Figure 7C. The numbers of the synapses correspond to the events described in Figure 7A and B as mono-, di-, and trisinaptic responses.

Discussion

This paper describes in detail the techniques required to isolate and maintain the adult guinea-pig brain *in vitro*. The long-term viability and suitability of this preparation for electrophysiological studies is demonstrated using the multisynaptic circuits of the primary visual pathway.

Perfused mammalian brain preparations

Since early in this century attempts have been made to isolate and perfuse the mammalian brain (Heymans and Heymans, 1926) for electrophysiological, biochemical, pharmacological or anatomical investigations. This was true originally for the brains of large species such as cats (Chute and Smyth, 1939; Suda *et al.*, 1963), dogs (Geiger and Magnus, 1947; Gilboe *et al.*, 1964, 1965), and monkeys (White *et al.*, 1963, 1964, 1966). Most such studies, however, were conducted *in situ* for a short period of time with a variable degree of vascular and neural isolation, and with little control of network viability at the electrophysiological level. For the most part, natural or artificial blood was circulated through the arterial system and quite extensive surgical procedures were involved.

In developing a completely isolated brain preparation, we chose to use rodents, specifically guinea-pigs, because their vertebral arteries

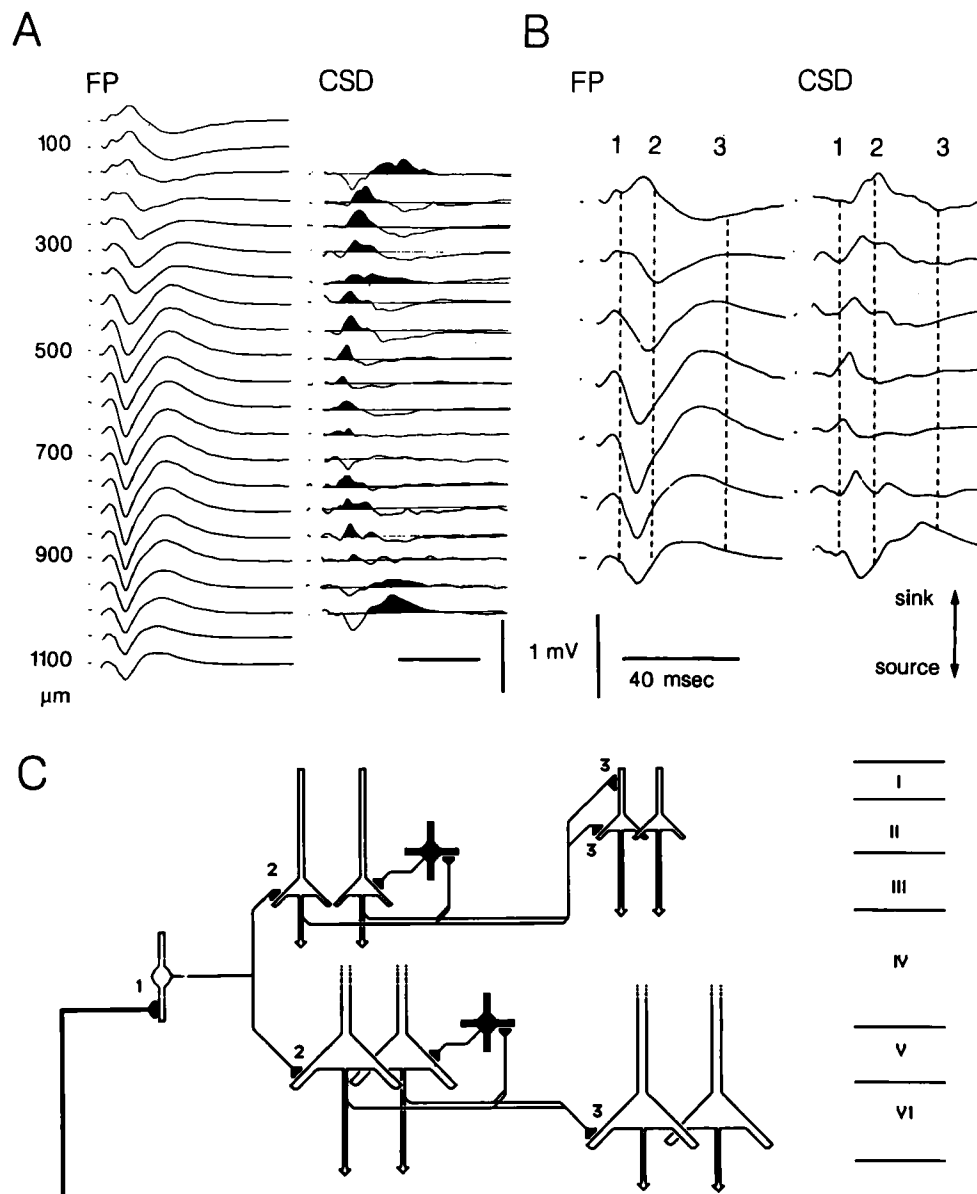


FIG. 7. Analysis of a field potential profile recorded from the visual cortex after optic tract stimulation. (A) Field potential profile (left column) and respective current source density (CSD) traces (right) along a penetration perpendicular to the cortical lamination. Sinks are illustrated in the CSD profile by black-filled areas. (B) Selected field potential (FP) and CSD traces in A at higher magnification. The broken lines mark the latencies of the synaptic events sequentially activated by optic tract stimulation. The numbers (1, 2 and 3) correspond to presumed mono-, di- and trisynaptic components, respectively, of the response. (C) Schematic drawing of the intracortical circuit activated by the optic tract stimulation. As for B, the numbers correspond to suggested synaptic components of the response.

are large compared to those of rats and mice. Originally, and until recently, many studies in rodents have used, with more or less success, the perfused *in situ* rat brain (Andjus *et al.*, 1967; Kriegstein *et al.*, 1972, 1973; Thompson *et al.*, 1968) or guinea-pig brain (Richerson and Getting, 1987, 1990). The obvious advantages of such *in situ* perfused preparations are that pharmacological experiments can be carried out that cannot be done *in vivo*. Also, unlike the completely isolated brain, studies on brainstem–spinal cord interactions, studies requiring the presence of sensory end organs, and studies of respiration requiring the preservation of respiratory muscles and peripheral feedback, can be implemented.

In contrast to these *in situ* perfused preparations, the *in vitro* approach using a completely isolated and perfused brain was deemed necessary for a number of reasons. The most important of all is that complete isolation *in vitro* allows, in addition to total control of the fluid environment, direct access to virtually any area of the brain, whether for stimulating or recording purposes. In addition, the examination of wide areas by optical imaging techniques is possible. An example of this increased access is shown in Figure 2D, where the brain is shown tilted, allowing stimulation or recording from various cranial nerves. Moreover, this type of preparation, when mastered, is not much more difficult to implement or maintain than are brain slices.

Interpretations of field potential and current source density analysis in visual pathways

Although several studies using intracellular and extracellular recordings have already been conducted in this preparation (see references in the Introduction) the present study focuses on the demonstration of the persistence of synaptic transmission in multisynaptic sensory pathways. Field potential analysis within the visual system rather than the study of single cell activity was utilized to evaluate the viability of complex circuits, because these potentials reflect more accurately the overall state of the system.

CSD analysis was implemented on averaged evoked potential profiles in order to evaluate the intrinsic connectivity of the superior colliculus, dLGN and visual cortex, and compare these to recordings made *in vivo*. The distribution and latencies of the sinks and sources derived from this analysis made it possible to infer the flow of synaptic excitation in the visual pathway. In the superior colliculus as reported in *in vivo* experiments in rats (Fukuda *et al.*, 1978), cats (Mitzdorf and Singer, 1977) and guinea-pigs (King and Johnson, 1985), the first wave (P1) evoked by optic tract stimulation corresponds to the Y-fibre activation of the distal dendrites of upper stratum griseum mediale (SGM) neurons. These dendrites are located between SGS and stratum opticum, 600–700 μm from the dorsal surface of the superior colliculus. The second wave (N2–P2) reflects activation of a set of these neurons by slow-conducting Y-fibre afferents. The late waves (N3 and N5) represent the activation induced by fast and slow W-fibre input to dendrites of SGS neurons whose somata are located near the border of the stratum opticum. As Freeman and Singer (1983) suggested, some polysynaptic component may contribute to the late portion of the evoked potential in the superior colliculus.

The dLGN in rodents, unlike that of primates and felines, does not have obvious laminations, although a 'hidden' organizational lamination of the retinal input to the thalamus was demonstrated in the rat (Reese, 1988). The segregation of retinal afferents to the rat dLGN is characterized by an outer shell receiving an X-fibre input and an inner core targeted by Y-fibres, while the W-fibre input projects mainly to the dorso-caudal portion of the dLGN (Mitzdorf and Singer, 1977). The electrophysiological data obtained in the isolated brain support a similar organization of optic tract input to the guinea-pig dLGN.

A detailed analysis of the AEPs evoked by optic tract stimulation has been reported for the cat visual cortex *in vivo* (Mitzdorf, 1985; Mitzdorf and Singer 1977) and *in vitro* (Bode-Greuel *et al.*, 1987). According to these authors, afferent activity evoked by optic tract stimulation is relayed in the visual cortex along two main pathways, both starting in layer IV and spreading to the supragranular and infragranular cortex. The supragranular pathway conveys activity from granular layer IV to layer III neurons, and through polysynaptic connections to layers II and I. The infragranular excitation is relayed sequentially to pyramidal neurons located in layers V and VI. A similar pattern of spatio-temporal synaptic activation was observed in our experiments when the optic tract or optic radiation was stimulated.

It is noteworthy that the latencies of the synaptic responses measured in the superior colliculus, dLGN and visual cortex were slower than those described *in vivo*. However, this is due to the lower temperature used *in vitro* as the latencies became quite close to those reported in *in vivo* studies when the temperature was raised to 33°C.

Ethical issues

Beyond the detailed technical procedures required to maintain an adult guinea-pig brain *in vitro*, the question of the ethical posture which must be taken regarding this preparation should be discussed.

If we assume, as we must, that the rest of an animal's body is primarily a supporting vessel for its brain—in the sense that it maintains the brain in the living condition and provides a means for the brain to interact with the external world—it follows *ipso facto* that the animal is, from an operational point of view, its brain. Therefore, the brain must be treated with the same concern and care as the animal in its integrity.

Two fundamental issues must then be treated. The first and most obvious relates to the pain and/or suffering the brain may experience under *in vitro* conditions. The second, more subtle, issue relates to the intrinsic dignity that the brain must command given its function.

The question of pain

The issue of whether an isolated brain is capable of experiencing some form of pain is probably the easier to address. Studies in various species including man have shown that acute severance of the connectivity between the brain and the external world does not lead to pain, other than that which may be inflicted during the moment of severance which, in our case, is done under anaesthesia. In quadriplegic patients, for example, the state of deafferentation resulting from a severed spinal cord is not painful in the acute stage. Phantom limb pains, which are a common outcome of such lesion, do occur, but only at a much later stage. Because the state of deafferentation *per se* does not lead to pain in the acute stage, one must ask whether direct manipulation of the brain mass itself results in pain. This is not the case since there are no pain receptors in the organ itself. This is attested by the lack of pain sensation during manipulation of the brain in human neurosurgery performed in totally awake individuals (Penfield and Jasper, 1954).

While the level of background activity is low in the cooled isolated brain (the EEG was indeed flat at that temperature), local anaesthesia must be applied topically to the peripheral stumps of the sensory cranial nerves, especially the trigeminal, if the brain is to be maintained at temperatures such that full activation of the encephalic mass is expected. However, if pain is deemed plausible or if the temperature is to be raised for sufficiently long periods, then systemic anaesthesia should be given.

The question of dignity

In maintaining an isolated brain *in vitro* it must be remembered, as previously stated, that the brain not only represents but is, to a great extent, the animal itself. It is required therefore that the type of electrophysiological and pharmacological procedures employed should take into account the fact that the brain is not an inanimate tissue, as other organs would be under *in vitro* condition. Rather, the paradigms to be utilized should follow those that would be employed under *in vivo* conditions. A 'naked' brain may not be viewed as embodying the perceptual abilities of the actual animal unless this philosophical question is constantly kept in mind. Otherwise, utilization of and research on the isolated brain could lead to improper treatment of this unique biological entity.

If these two major categories of problems are kept carefully in mind, studies using the isolated brain preparation can provide significant scientific insights. For it presents the possibility of scientific analysis of every aspect of neuroscience from the molecular to the higher brain functions—utilizing invasive as well as non-invasive techniques. Thus it may prove to be an invaluable tool for the study of normal and pathological brain function.

Acknowledgements

This research was supported by the US National Institutes of Health (NINDS grants NS13742 and NS07124 to R. L.), the Swiss National Science Foundation

(grants 3.288.0.85 and 3.560.0.86 to M. M.) and Italian National Research Council (to M. De C.). We thank F. Strub, R. Schumacher and R. Golder for photography and artwork.

Abbreviations

AEP	averaged evoked potential
CNS	central nervous system
CSD	current source density
dLGN	dorsal lateral geniculate nucleus
HEPES	<i>N</i> -(2-hydroxyethyl)piperazine- <i>N'</i> -(2-ethanesulphonic acid)
SGM	stratum griseum mediale
SGP	stratum griseum profundum
SGS	stratum griseum superficialis

References

- Alonso, A., de Curtis, M. and Llinás, R. (1990) Postsynaptic Hebbian and non-Hebbian long-term potentiation of synaptic efficacy in the entorhinal cortex in slices and in the isolated adult guinea-pig brain. *Proc. Natl Acad. Sci. USA*, **87**, 9280–9284.
- Andjus, R. K., Suhara, K. and Sloviter, H. A. (1967) An isolated perfused rat brain preparation, its spontaneous and stimulated activity. *J. Appl. Physiol.*, **22**, 1033–1039.
- Bagust, J., Kelly, M. M. and Kerkut, G. A. (1985) An isolated mammalian brainstem spinal cord preparation suitable for the investigation of descending control of motor activity. *Brain Res.*, **327**, 370–374.
- Bode-Greuel, K. M., Singer, W. and Aldenhoff, J. B. (1987) A current source density analysis of field potentials evoked in slices of visual cortex. *Exp. Brain Res.*, **69**, 213–219.
- Bourque, C. W. and Renaud, L. P. (1984) Activity patterns and osmosensitivity of rat superoptic neurones in perfused hypothalamic explants. *J. Physiol. (Lond.)*, **349**, 631–642.
- Chute, A. L. and Smyth, D. H. (1939) Metabolism of the isolated perfused cat's brain. *Q. J. Exp. Physiol.*, **29**, 379–394.
- De Curtis, M. and Llinás, R. R. (1989) Spontaneous and evoked cortical activity in the isolated adult guinea pig brain maintained in vitro. *Soc. Neurosci. Abstr.*, **15**, 660.
- De Curtis, M., Paré, D. and Llinás, R. (1991) The electrophysiology of the olfactory hippocampal circuit in the isolated and perfused adult mammalian brain. *Hippocampus*, **1**, 341–354.
- Freeman, B. and Singer, W. (1983) The direct and indirect visual inputs to the superficial layers of the cat superior colliculus; a current source density analysis of electrically evoked potentials. *J. Neurophysiol.*, **49**, 1075–1091.
- Freeman, J. A. and Nicholson, C. (1975) Experimental optimization of current source density technique for anuran cerebellum. *J. Neurophysiol.*, **49**, 369–382.
- Fukuda, Y., Susuki, D. A. and Iwama, K. (1978) Characteristics of optic nerve innervation in the rat superior colliculus as revealed by field potential analysis. *Jpn. J. Physiol.*, **28**, 347–365.
- Fulton, B. P. and Walton, K. (1986) Electrophysiological properties of neonatal rat motoneurons studied *in-vitro*. *J. Physiol. (Lond.)*, **370**, 651–680.
- Geiger, A. and Magnus, J. (1947) The isolation of the cerebral circulation and the perfusion of the brain of the living cat. *Am. J. Physiol.*, **149**, 517–537.
- Gilboe, D. D., Cotanch, W. W. and Glover, M. B. (1964) Extracorporeal perfusion of the isolated head of a dog. *Nature*, **202**, 399–400.
- Gilboe, D. D., Cotanch, W. W. and Glover, M. B. (1965) Isolation and mechanical maintenance of the dog brain. *Nature*, **206**, 94–96.
- Grantyn, R. and Kertman, H. (1992) *Electrophysiological Methods for In Vitro Studies in Vertebrate Neurobiology*. Wiley-Liss, New York.
- Hackett, J. T. and Llinás, R. (1973) Functional properties of frog cerebellar circuit *in vitro*: Blockade by magnesium. *Fed. Proc.*, **32**, 1241.
- Heymans, J. F. and Heymans, C. (1926) Recherches physiologiques et pharmacodynamiques sur la tête isolée du chien. II. Sur l'influence respiratoire et pneumogastrique de l'adrénaline, de l'atropine, de la pituitrine, de la nicotine et des digitaliques. *Arch. Int. Pharmacodyn.*, **32**, 9–41.
- Kay, A. R. and Wong, R. K. S. (1986) Isolation of neurons suitable for patch-clamping from adult mammalian central nervous systems. *J. Neurosci. Methods*, **6**, 227–238.
- Khatieb, A., Serafin, M. and Mühlethaler, M. (1990) Intracellular recording from tubermammillary neurons in an isolated and perfused whole brain of guinea pig *in vitro*. *Soc. Neurosci. Abstr.*, **16**, 1255.
- King, A. J. and Johnson, M. S. (1985) The synaptic distribution of the retinal input in the superficial layers of the guinea-pig superior colliculus. *Proc. R. Soc. Lond. Ser. B*, **225**, 129–146.
- Kriegstein, G., Kriegstein, J. and Urban, W. (1972) Long survival time of an isolated perfused rat brain. *J. Neurochem.*, **19**, 885–886.
- Kriegstein, J., Stock, R. and Geiger, H. (1973) Influence of therapeutic and toxic doses of neuroleptics and antidepressants on energy metabolism of the isolated perfused rat brain. *Naunyn-Schmiedeberg's Arch. Pharmacol.*, **279**, 243–264.
- Llinás, R. and Mühlethaler, M. (1988a) An electrophysiological study of the *in vitro* perfused brain stem–cerebellum of adult guinea-pig. *J. Physiol. (Lond.)*, **404**, 215–240.
- Llinás, R. and Mühlethaler, M. (1988b) Electrophysiology of guinea-pig cerebellar nuclear cells in the *in vitro* brainstem–cerebellar preparation. *J. Physiol. (Lond.)*, **404**, 241–258.
- Llinás, R., Yarom, Y. and Sugimori, M. (1981) Isolated mammalian brain *in vitro*: new technique for analysis of electrical activity of neuronal circuit function. *Fed. Proc.*, **40**, 2242–2245.
- Llinás, R., Mühlethaler, M. and Walton, K. (1989) Electrophysiology of the adult guinea pig brain *in vitro*. *J. Physiol. (Lond.)*, **414**, 16P.
- Mitzdorf, U. (1985) Current source density method and application in the cat cerebral cortex: investigation of evoked potentials and EEG phenomena. *Physiol. Rev.*, **65**, 37–100.
- Mitzdorf, U. and Singer, W. (1977) Laminar segregation of afferents to lateral geniculate nucleus of the cat: an analysis of current source density. *J. Neurophysiol.*, **40**, 1227–1244.
- Mitzdorf, U. and Singer, W. (1978) Prominent excitatory pathways in the cat visual cortex (A17 and A18): a current source density analysis of electrically evoked potentials. *Exp. Brain Res.*, **33**, 371–394.
- Mühlethaler, M. and MacVicar, B. A. (1989) Synchronization of seizure activity in the hippocampus and entorhinal cortex of the perfused whole guinea-pig brain. *Soc. Neurosci. Abstr.*, **15**, 1031.
- Mühlethaler, M. and Serafin, M. (1990) Thalamic spindles in an isolated and perfused brain preparation *in vitro*. *Brain Res.*, **524**, 17–21.
- Mühlethaler, M., Serafin, M. and Vidal, P. P. (1987) Electrophysiological properties of guinea-pig facial motoneurons: combined study in the acute preparation and the *in vitro* whole brain. *Soc. Neurosci. Abstr.*, **13**, 853.
- Mühlethaler, M., Cand, P., Henauer, R. and Serafin, M. (1990) Design of a microprocessor controlled thermoregulated chamber for perfusing *in vitro* mammalian brain preparations. *J. Neurosci. Methods*, **31**, 215–223.
- Nicholson, C. and Freeman, J. A. (1975) Theory of current source density analysis and determination of conductive tensor for anuran cerebellum. *J. Neurophysiol.*, **38**, 356–368.
- Otsuka, M. and Konishi, S. (1974) Electrophysiology of mammalian spinal cord *in vitro*. *Nature*, **252**, 733–734.
- Paré, D. and Llinás, R. (1991) Spatiotemporal organization of the entorhinal input to the hippocampus: Multiple electrode recordings from the adult guinea pig brain maintained *in vitro*. *Soc. Neurosci. Abstr.*, **17**, 128.
- Paré, D., de Curtis, M. and Llinás, R. (1992) Role of the hippocampal–entorhinal loop in temporal lobe epilepsy: extra- and intracellular study in the isolated guinea pig brain *in vitro*. *J. Neurosci.*, **12**, 1867–1881.
- Penfield, W. and Jasper, H. (1954) *Epilepsy and the Functional Anatomy of the Human Brain*. Little, Brown and Company, Boston.
- Reese, B. E. (1988) 'Hidden lamination' in the dorsal lateral geniculate nucleus: the functional organization of this thalamic region in the rat. *Brain Res. Rev.*, **13**, 119–137.
- Rice, M. E. and Nicholson, C. (1991) Diffusion characteristics and extracellular volume fraction during normoxia and hypoxia in slices of rat neocortex. *J. Neurophysiol.*, **65**, 264–272.
- Richerson, G. B. and Getting, P. A. (1987) Maintenance of complex neuronal function during perfusion of the mammalian brain. *Brain Res.*, **409**, 128–132.
- Richerson, G. B. and Getting, P. A. (1990) Preservation of integrity functions in a perfused guinea-pig brain. *Brain Res.*, **517**, 7–18.
- Serafin, M. and Mühlethaler, M. (1988) Electrophysiological study of reticular neurons in an isolated whole brain of guinea-pig and in brainstem slices. *Experientia*, **44**, A70.
- Suda, I., Adachi, C. and Kito, K. (1963) Studies on the isolated cat's brain *in vitro*. I. Isolation methods and observation on the spontaneous electrical activity. *Kobe J. Med. Sci.*, **9**, 41–67.
- Thompson, A. M., Robertson, R. C. and Bauer, T. A. (1968) A rat head perfusion technique developed for the study of brain uptake of materials. *J. Appl. Physiol.*, **24**, 407–411.
- Walton, K. and Llinás, R. (1982) Electrophysiological activity recorded from the cerebral cortex of an isolated mammalian brain maintained *in vitro*. *Soc. Neurosci. Abstr.*, **8**, 935.

926 Isolated guinea-pig brain

- White, R. J., Albin, M. S. and Verdura, J. (1963) Isolation of the monkey brain: *in vitro* preparation and maintenance. *Science*, **141**, 1060–1061.
- White, R. J., Albin, M. S. and Verdura, J. (1964) Preservation of viability in the isolated monkey brain utilizing a mechanical extracorporeal circulation.

Nature, **202**, 1082–1083.

- White, R. J., Albin, M. S., Verdura, J. and Locke, G. E. (1966) Prolonged whole-brain refrigeration with electrical and metabolic recovery. *Nature*, **209**, 1320–1322.